LumaChart

THE HOSPITAL EHR BUILT FOR PUBLIC HEALTH

Redesigning the medical record around health & the healer.

A final planning document for an evidence-based electronic health record that prevents clinician burnout, organizes care around prevention and healthspan, and turns consented data into public-health knowledge.

LUMAEHR.com

Product Design Brief · v1.0 Grounded in 24 peer-reviewed sources · 22 verified PubMed IDs **Prepared** July 2026

EXECUTIVE SUMMARY

A third choice — architected around well-being, not billing.

Epic and Oracle Health (Cerner) were built around documentation and reimbursement. The clinical evidence is now unambiguous that this architecture is a leading, measurable driver of a national burnout epidemic — and that the same billing-first logic crowds out prevention for patients. **LumaChart** is a clean-sheet electronic health record built around three commitments the incumbents treat as afterthoughts.

1 **Clinician sustainability**

Burnout prevention is engineered *into* the workflow — lean documentation decoupled from billing, delegated clerical work, and **Canary**, a private early-warning system that reads a clinician's own EHR-usage signals and intervenes before exhaustion sets in.

2 **Patient healthspan**

The record is organized around **health maintenance** — evidence-based prevention, screening, and longevity — rather than billing codes, with every recommendation traceable to the literature.

3 **Public-health research**

Leaner, structured data plus explicit patient consent makes prospective, population-scale research a first-class feature — exactly what the National Academy of Medicine says the field urgently needs.

> "We're spending our days doing the wrong work… we are disconnected from our purpose and have lost touch with the things that give joy and meaning to our work." **— Christine Sinsky, MD, AMA** PMID 29365301

This document reviews the evidence, defines the product, and maps **every feature to a specific, verifiable source.** It is a design brief and interactive-prototype specification — not a certified production system (see *Responsible Design*, p.10).

WHY THE EHR MUST BE REDESIGNED

The record became the workload.

Time-motion and EHR event-log studies quantify what clinicians feel: the modern EHR converted highly trained physicians into data-entry clerks and moved hours of unpaid work into the evening.

5.9 hr
of an 11.4-hour workday spent in the EHR, per primary-care physician PMID 28893811

~2:1
EHR & desk work for every 1 hour of direct patient care PMID 27595430

1.4 hr
of after-hours "pajama time" in the EHR every day PMID 28893811

44%
of EHR time is clerical (documentation, orders, billing, security) PMID 28893811

4×
longer U.S. notes than the same EHR abroad — driven by billing rules, not medicine PMID 29801050

>50%
of U.S. physicians report substantial burnout — ~2× other workers PMID 26653297

! The human toll

- Burnout = emotional exhaustion + depersonalization + low personal accomplishment (Maslach). PMID 29365301
- +25% odds of alcohol misuse; **+200% odds of suicidal ideation.** NAM 2017
- Independently predicts major medical errors & malpractice suits. NAM 2017
- Each 1-pt rise in exhaustion → 28% higher odds of cutting clinical effort. NAM 2017

$ The system toll

- Burned-out physicians are **2× as likely to leave;** replacement costs **$500K–$1M** each. PMID 29365301
- Front-line fields — family & internal medicine, EM, neurology — are highest risk. NAM 2017
- Lost productivity ≈ eliminating the graduating classes of **seven medical schools.** NAM 2017

> "The highly trained U.S. physician has become a data-entry clerk… we believe the regulations around documentation and billing likely play a larger role." **— Downing, Bates & Longhurst**, *Ann Intern Med* 2018 PMID 29801050

THREE DESIGN PRINCIPLES

Reimagine the work — don't digitize the paperwork.

The evidence points to solutions, not just problems. LumaChart is built on three principles drawn directly from the literature.

1 · The Quadruple Aim is the product spec

Better population health, better experience of care, and lower cost are unreachable without a fourth aim: **care of the provider.** LumaChart treats clinician well-being as a primary design constraint, not a wellness webpage. PMID 25384822

2 · Separate the clinical note from the cash register

U.S. notes bloated to 4× international length because billing and compliance data were poured into the clinical narrative. LumaChart keeps the note **clinically lean** and routes coding/compliance to a separate, mostly-automated layer — the single highest-leverage documentation fix in the literature. PMID 29801050 PMID 24842418

3 · Measure well-being, because you manage what you measure

Mayo Clinic benchmarks physician well-being annually and puts it on the CEO scorecard. LumaChart builds validated measurement in — the **MBI** gold standard and **ProQOL** — surfaced privately to clinicians and in aggregate to leaders. NAM 2017 PMID 28347144

The unifying idea

Clinician sustainability and patient healthspan are the same coin: a record optimized for **health and well-being** instead of **billing.** Fixing the clinician's day (leaner notes, delegated clerical work, team-based care) is the *same* act as reorienting the patient's record toward prevention. That is what makes LumaChart a genuine third choice rather than a reskin.

Interventions work: meta-analyses of dozens of trials find that both **organizational** and **individual** approaches measurably reduce burnout. LumaChart deliberately does both. PMID 27692469 PMID 27918798

CLINICIAN EXPERIENCE

A workspace that protects the person using it.

Lean chart & documentation

- Clinical note decoupled from billing/compliance.
- Ambient / voice capture; auto problem-oriented summaries.
- Pre-visit patient questionnaire flows into the note. PMID 29132154

Team "care choreography"

- Role-based delegation; standing orders; shared inbox.
- MA/scribe-assisted documentation (RCT: ↑ satisfaction). PMID 28893812
- Modeled on APEX: burnout **53%→13%** in 6 months. PMID 29365301

Time Well Spent

- Live session timer; face-time vs desktop-time.
- After-hours / pajama-time trend, inbox-burden meter. PMID 28893811

Wellness Center

- Guided meditation & breathing micro-breaks. PMID 18954193
- 8-week mindfulness track (evidence: gray-matter change). PMID 21071182
- Gratitude practice surfacing patient thank-you notes.

Canary — the burnout early-warning engine

The flagship feature, originating in the author's own "Canary" concept: like a canary in a coal mine, it gives an early warning. Canary continuously reads a clinician's **own** event-log signals — session length, after-hours load, inbox backlog, documentation minutes — against their personal baseline, and escalates gently:

micro-break nudge · **mindfulness prompt** · **extended-login warning** · **"trending toward burnout risk"**

Designed as support, not surveillance. Canary is private to the clinician by default and non-punitive; only de-identified, aggregate signals reach wellness programs. Any predictive threshold must be clinically validated before real-world use. Signal set is grounded in the measured drivers of burnout. PMID 28893811 PMID 27313121

PATIENT EXPERIENCE · THEME: HEALTH MAINTENANCE

A record organized around living longer & better.

The patient portal is not a billing statement with a message box. Its organizing logic is **healthspan**: what keeps you well, what's due, and what the evidence says it buys you.

Healthspan home

- Prevention status at a glance; what's due now.
- Lifestyle vitals & a plain-language healthspan summary.

Prevention & screening schedule

- Age/sex-appropriate screening & vaccines.
- Each item carries an evidence card + real citation.
- Framed as "why this matters," not "claim submitted."

Longevity tracker

- Modifiable risks: BP, A1c, lipids, activity, sleep, tobacco.
- Each tied to evidence-based guidance the care team set.

Structured, consistent counseling

- Guideline-consistent scripts prevent the contradictory, "improvised" counseling shown to erode patient trust. PMID 25208096

Prevention also lightens the clinician: the APEX model succeeded precisely because prevention (vaccination, cancer screening) was systematized by the team rather than improvised in the room — improving screening rates *and* cutting burnout at the same time. Patient healthspan and clinician sustainability reinforce each other. PMID 29365301

> Health maintenance is the theme — the portal answers "how do I stay well and live longer?" before it ever answers "what do I owe?"

PUBLIC-HEALTH RESEARCH

Consented data that gives back to the population.

The NAM explicitly calls for **prospective, longitudinal research** linking clinician well-being and care delivery to patient outcomes — and notes leaner documentation "would enable better use for research." LumaChart makes that a native capability. PMID 29801050

✔ Patient consent layer

- Explicit, revocable opt-in to contribute de-identified data.
- Plain-language transparency on exactly what is shared.

◫ Researcher console

- Consented cohort counts; an example prospective study.
- Privacy guardrails; well-being ↔ outcomes linkage the NAM asks for.

INTERFACE AS CLINICAL INTERVENTION

Restore Mode — eyestrain reduction as a feature.

Physician strain includes **eye strain**. LumaChart ships a warm, low-glare dark interface — **"Restore Mode"** — as the default and names it as a wellbeing feature, with a one-tap switch to familiar looks for clinicians who want them.

☾ Restore (default)

Dark, warm, low-glare; reduced motion; generous spacing to lower visual & cognitive load on long shifts.

Classic & Modern Clinical

Conventional light themes for clinicians who prefer the familiar Epic/Cerner density. The clinician stays in control.

EVIDENCE → FEATURE

Every feature traces to a source.

FINDING IN THE LITERATURE	LUMACHART FEATURE	SOURCE
1.4 hr/day after-hours; ~2:1 desk-to-patient ratio	Canary early-warning; extended-login warning; Time Well Spent	28893811 27595430
Inbox = 24%, clerical = 44% of EHR time	Inbox-burden meter; shared/team inbox triage	28893811
U.S. notes 4× too long; billing is the real cause	Lean note decoupled from billing/compliance layer	29801050 24842418
Scribes ↑ satisfaction (RCT); APEX 53%→13%	Team choreography; MA/scribe-assisted documentation	28893812 29365301
Patients contributing to notes ↑ engagement	Pre-visit questionnaire feeding the chart	29132154
Care of the provider = the 4th aim	Well-being as a primary design constraint	25384822
"You manage what you measure"; MBI / ProQOL	Wellness Gauge; validated self-checks	NAM 2017 · 28347144
Loving-kindness & 8-wk mindfulness change affect/brain	Wellness Center meditations & micro-breaks	18954193 21071182
Org + individual interventions both reduce burnout	Combined workflow (org) + Wellness (individual) design	27692469 27918798
EHR clerical burden turns MDs into "data-entry clerks"	Reimagined workflow, not digitized paper	29133547
Leaner data "enables better use for research"; NAM call	Consented prospective research layer	29801050 · NAM 2017
Contradictory, improvised counseling erodes trust	Structured, guideline-consistent counseling module	25208096
Physician strain incl. eye strain; clinician control	Restore (dark) default + Classic/Modern toggle	Design principle

WHAT THIS IS — AND WHAT IT IS NOT

An honest path from prototype to production.

LumaChart v1 is an **evidence-grounded vision and interactive prototype**. It demonstrates the concept end-to-end; it is not yet a certified, deployable hospital system. Naming that gap plainly is part of the design.

✓ What the prototype delivers

- Clickable clinician, patient & researcher experiences.
- Working Canary demo (accelerated timer).
- Every clinical claim cited to a verifiable source.

→ Gates before real-world use

- **ONC Health IT certification** & interoperability (FHIR).
- **HIPAA** security review & independent audit.
- **IRB approval** for the research layer.
- **Clinical validation** of any Canary predictive threshold.

On Canary and monitoring. Any system that reads clinician activity can be misused as surveillance. LumaChart's design commitments — clinician-owned data, private-by-default, aggregate-only for management, non-punitive framing, and no automated employment consequences — are load-bearing, not optional. They are what separate a wellbeing tool from a productivity dragnet.

Evidence integrity. This document cites 24 sources; 22 carry PubMed IDs verified directly against PubMed's E-utilities during preparation. No identifier here was fabricated or assumed. Two items (the NAM discussion paper and the author's MPH capstone) are not PubMed-indexed and are cited in full.

PROTOTYPE SCOPE & SEQUENCE

What we build now.

A single self-contained web application with a role switcher (Clinician · Patient · Researcher) and the three-theme system, delivered in six increments.

1 · App shell & theming. Role switcher, Restore/Classic/Modern themes, global session timer, LumaChart identity.

2 · Canary engine. Event-log signals vs baseline, escalating nudges, extended-login warning (accelerated demo clock).

3 · Clinician workspace. Dashboard (panel + inbox-burden), lean chart, Wellness Center, Time Well Spent, Wellness Gauge.

4 · Patient portal. Healthspan home, prevention/screening schedule with citations, longevity tracker, pre-visit questionnaire.

5 · Research layer. Patient consent panel; researcher console with cohort counts & guardrails.

6 · QA. Verify rendering & interaction across every screen and all three themes; confirm Canary fires and citations display.

Definition of done (prototype)

A reviewer can open one file, switch roles and themes, watch Canary escalate to an extended-login warning, read a patient healthspan plan with live citations, and see the consented-research console — all without a network connection.

Stack: dependency-free HTML/CSS/JavaScript — no build step, opens in any browser. This keeps the prototype inspectable, portable, and honest about being a front-end demonstration rather than a clinical back end.

EVIDENCE BASE · 22 VERIFIED PUBMED IDS

References

1. Arndt BG, Beasley JW, Watkinson MD, et al. Tethered to the EHR: primary care physician workload assessment using EHR event log data and time-motion observations. *Ann Fam Med.* 2017;15(5):419–426. **PMID 28893811**
2. Sinsky C, Colligan L, Li L, et al. Allocation of physician time in ambulatory practice: a time and motion study in 4 specialties. *Ann Intern Med.* 2016;165(11):753–760. **PMID 27595430**
3. Tai-Seale M, Olson CW, Li J, et al. Electronic health record logs indicate that physicians split time evenly between seeing patients and desktop medicine. *Health Aff (Millwood).* 2017;36(4):655–662. **PMID 28373331**
4. Downing NL, Bates DW, Longhurst CA. Physician burnout in the electronic health record era: are we ignoring the real cause? *Ann Intern Med.* 2018;169(1):50–51. **PMID 29801050**
5. Collier R. Electronic health records contributing to physician burnout. *CMAJ.* 2017;189(45):E1405–E1406. **PMID 29133547**
6. Shanafelt TD, Dyrbye LN, Sinsky C, et al. Relationship between clerical burden and characteristics of the electronic environment with physician burnout and professional satisfaction. *Mayo Clin Proc.* 2016;91(7):836–848. **PMID 27313121**
7. Zulman DM, Shah NH, Verghese A. Evolutionary pressures on the electronic health record: caring for complexity. *JAMA.* 2016;316(9):923–924. **PMID 27532804**
8. Dyrbye LN, Shanafelt TD, Sinsky CA, et al. Burnout among health care professionals: a call to explore and address this underrecognized threat to safe, high-quality care. *NAM Perspectives.* Discussion Paper, National Academy of Medicine, Washington, DC; 2017. doi:10.31478/201707b
9. Shanafelt TD, Hasan O, Dyrbye LN, et al. Changes in burnout and satisfaction with work-life balance in physicians and the general US working population between 2011 and 2014. *Mayo Clin Proc.* 2015;90(12):1600–1613. **PMID 26653297**
10. Shanafelt TD, Boone S, Tan L, et al. Burnout and satisfaction with work-life balance among US physicians relative to the general US population. *Arch Intern Med.* 2012;172(18):1377–1385. **PMID 22911330**
11. Wright AA, Katz IT. Beyond burnout — redesigning care to restore meaning and sanity for physicians. *N Engl J Med.* 2018;378(4):309–311. **PMID 29365301**
12. Bodenheimer T, Sinsky C. From triple to quadruple aim: care of the patient requires care of the provider. *Ann Fam Med.* 2014;12(6):573–576. **PMID 25384822**
13. West CP, Dyrbye LN, Erwin PJ, Shanafelt TD. Interventions to prevent and reduce physician burnout: a systematic review and meta-analysis. *Lancet.* 2016;388(10057):2272–2281. **PMID 27692469**
14. Panagioti M, Panagopoulou E, Bower P, et al. Controlled interventions to reduce burnout in physicians: a systematic review and meta-analysis. *JAMA Intern Med.* 2017;177(2):195–205. **PMID 27918798**
15. Shanafelt TD, Noseworthy JH. Executive leadership and physician well-being: nine organizational strategies to promote engagement and reduce burnout. *Mayo Clin Proc.* 2017;92(1):129–146. **PMID 27871627**
16. Gidwani R, Nguyen C, Kofoed A, et al. Impact of scribes on physician satisfaction, patient satisfaction, and charting efficiency: a randomized controlled trial. *Ann Fam Med.* 2017;15(5):427–433. **PMID 28893812**
17. Mafi JN, Gerard M, Chimowitz H, et al. Patients contributing to their doctors' notes: insights from expert interviews. *Ann Intern Med.* 2018;168(4):302–305. **PMID 29132154**
18. Sinsky CA, Beasley JW, Simmons GE, Baron RJ. Electronic health records: design, implementation, and policy for higher-value primary care. *Ann Intern Med.* 2014;160(10):727–728. **PMID 24842418**
19. Fredrickson BL, Cohn MA, Coffey KA, Pek J, Finkel SM. Open hearts build lives: positive emotions, induced through loving-kindness meditation, build consequential personal resources. *J Pers Soc Psychol.* 2008;95(5):1045–1062. **PMID 18954193**
20. Hölzel BK, Carmody J, Vangel M, et al. Mindfulness practice leads to increases in regional brain gray matter density. *Psychiatry Res.* 2011;191(1):36–43. **PMID 21071182**
21. O'Mahony S, Ziadni M, Hoerger M, Levine S, Baron A, Gerhart J. Compassion fatigue among palliative care clinicians: findings on personality factors and years of service. *Am J Hosp Palliat Care.* 2018;35(2):343–347. **PMID 28347144**
22. Tuthill EL, Chan J, Butler LM. Challenges faced by healthcare providers offering infant feeding counseling to HIV-positive women in sub-Saharan Africa: a review of current research. *AIDS Care.* 2015;27(1):17–24. **PMID 25208096**
23. Sinsky CA, Willard-Grace R, Schutzbank AM, et al. In search of joy in practice: a report of 23 high-functioning primary care practices. *Ann Fam Med.* 2013;11(3):272–278. **PMID 23690328**
24. McAhren LG, McAneny BL, Padula WV. Toward an economy of resilience: rates of physician burnout & rates of physician employment in the workforce. MPH Capstone, Johns Hopkins Bloomberg School of Public Health; 2018.